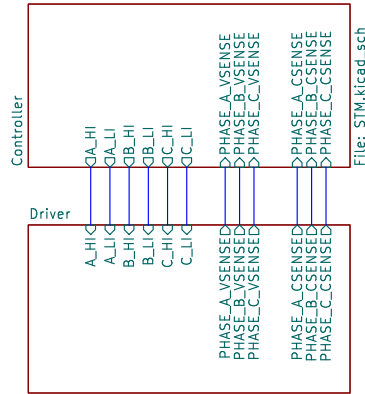


Schematic



File: DRV8313.kicad_sch

12V TO 5V

File: TPS54334DDA.kicad_sch

Mekatra SastaDrive

3-Phase FOC Motor Driver

SAFETY CHECK

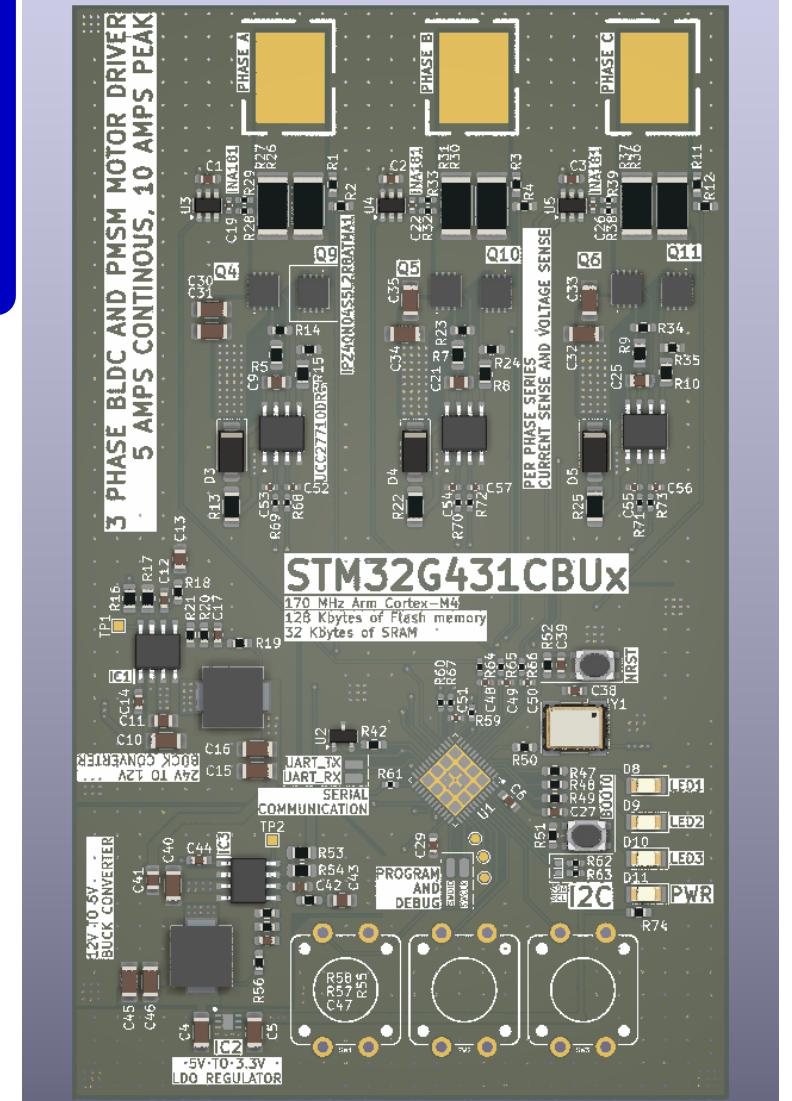
24V / 5A CONT / 10A PEAK
! VIN: 18V MIN 24V NOM 26V MAX
! DO NOT REVERSE POLARITY
! NO ONBOARD FUSE - FUSE EXTERNALLY
DISCHARGE CAPS BEFORE HANDLING
FLOATING PHASE = HIGH VOLTAGE
DO NOT TOUCH WHEN ACTIVE
DISCONNECT POWER BEFORE WIRING
DIRECTION: SWAP ANY 2 PHASES IN FW
! HIGH VOLTAGE SWITCHING ZONE
! DO NOT PROBE PHASE NODE LIVE
! VCC = 24V - DO NOT SHORT
MOSFET: IPZ40N04 40V 40A
HOT SURFACE DURING OPERATION
ISENSE: 10mR / GAIN 20 / 5A FS
CALIBRATE OFFSET AT EVERY BOOT
ADC SAT AT >8.25A BUS CURRENT
3.3V LOGIC - DO NOT APPLY 5V
STM32G431CBT6 | 3.3V ONLY
SWD: SWDIO SWDCLK 3V3 GND
UART: TX RX GND
! GPIO MAX 3.3V - NO 5V SIGNALS
BOOT0 LOW = NORMAL HIGH = DFU

Overview

SastaDrive is a compact, low-cost, 3-phase Field-Oriented Control (FOC) capable motor driver designed for custom BLDC and PMSM motors. It integrates power stage, gate drive, current sensing, bus voltage sensing, and an STM32G431CBT6 microcontroller on a 2-layer PCB. The board is designed for RnD and small-volume applications requiring smooth, precise torque delivery at low to mid RPM.

Key Features

- 24 V nominal DC bus input, 3-phase BLDC/PMSM output
- FOC-capable with 3-phase inline current sensing (INA181A2 + 20 mOhm shunts)
- STM32G431CBT6 MCU with hardware complementary PWM, dual simultaneous ADC
- UCC27710 half-bridge gate drivers with bootstrap supply
- 24 V to 12 V onboard buck converter (TPS54334)
- I2C encoder header for AS5600 magnetic encoder module
- SWD debug header + UART header for firmware development
- Aluminium frame compatible - bottom copper pour acts as heatsink interface

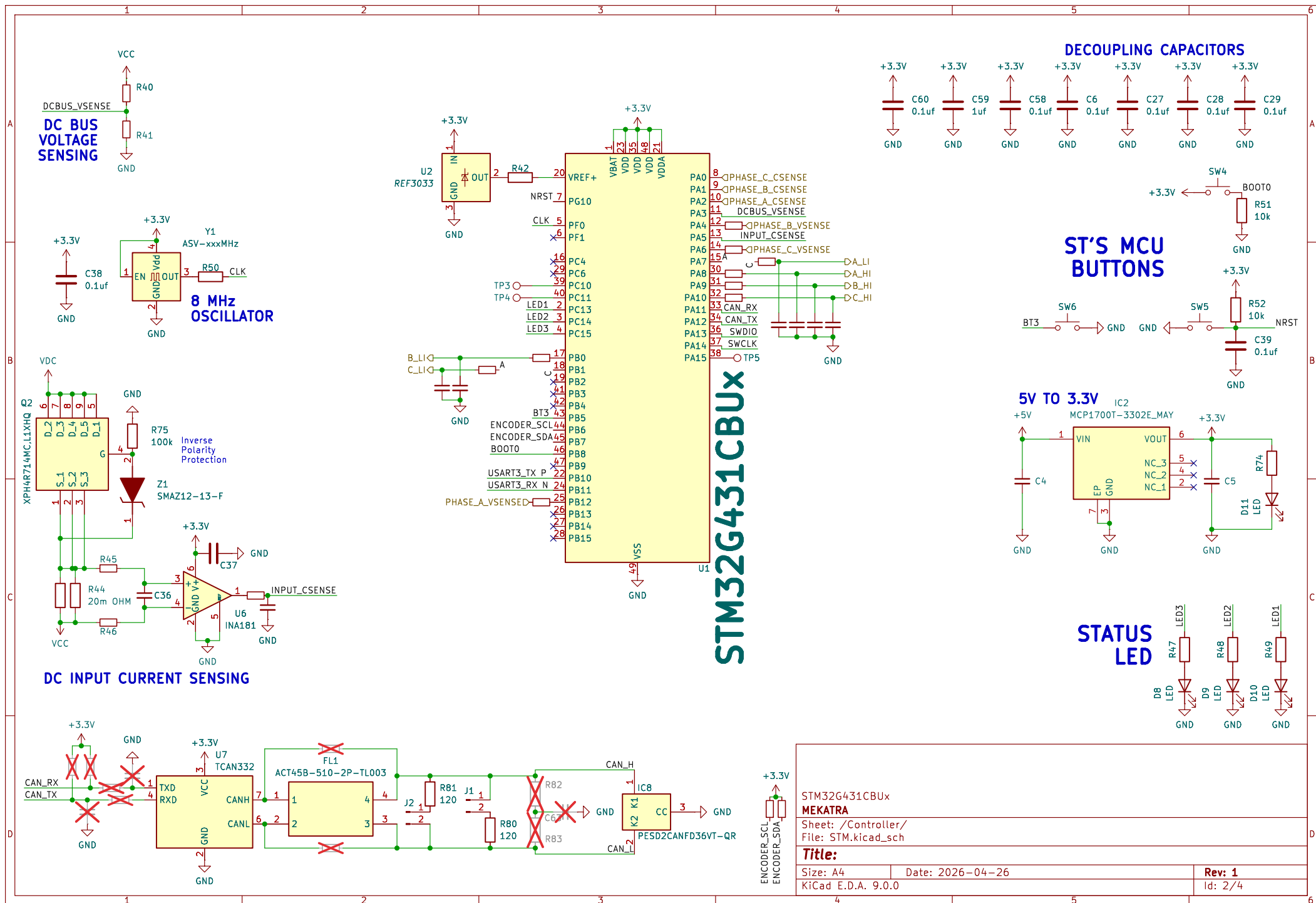


SastaDrive Schematic
3-Phase FOC Motor Driver
MEKATRA
Sheet: /
File: ESC Drive.kicad_sch

Title:

Size: A4 Date: 2026-04-26
KiCad E.D.A. 9.0.0

Rev: 1
Id: 1/4



STM32G431CBUX

MEKATRA

Sheet: /Controller/

File: STM.kicad_sch

Title:

Size: A4

Date: 2026-04-26

KiCad E.D.A. 9.0.0

Rev: 1

Id: 2/4

PHASE A

PHASE C

PHASE B

24V TO 12V
BUCK CONVERTER

TPS54334DDA BUCK CONVERTER
UCC27710DR GATE DRIVER IC
IPZ40N04S5L2R8ATMA1 MOSFETS
MEKATRA

Sheet: /Driver/
File: DRV8313.kicad_sch

Title: **ACTIVE_INVERTER**

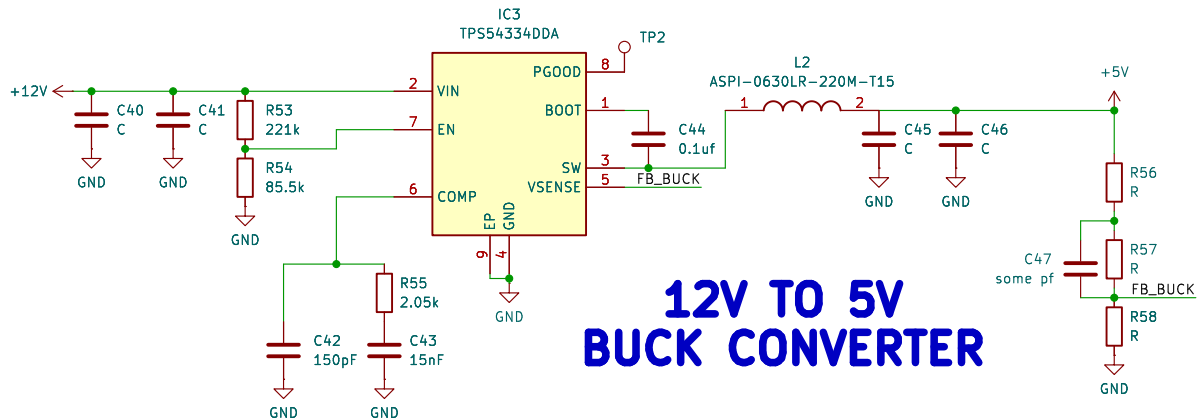
Size: A3

Date: 2026-04-26

Rev: **1**

KiCad E.D.A. 9.0.0

Id: 3/4



TPS54334DDA BUCK CONVERTER

MEKATRA

Sheet: /12V TO 5V/

File: TPS54334DDA.kicad_sch

Title:

Size: A5

Date: 2026-04-26

Rev: 1

KiCad E.D.A. 9.0.0

Id: 4/4